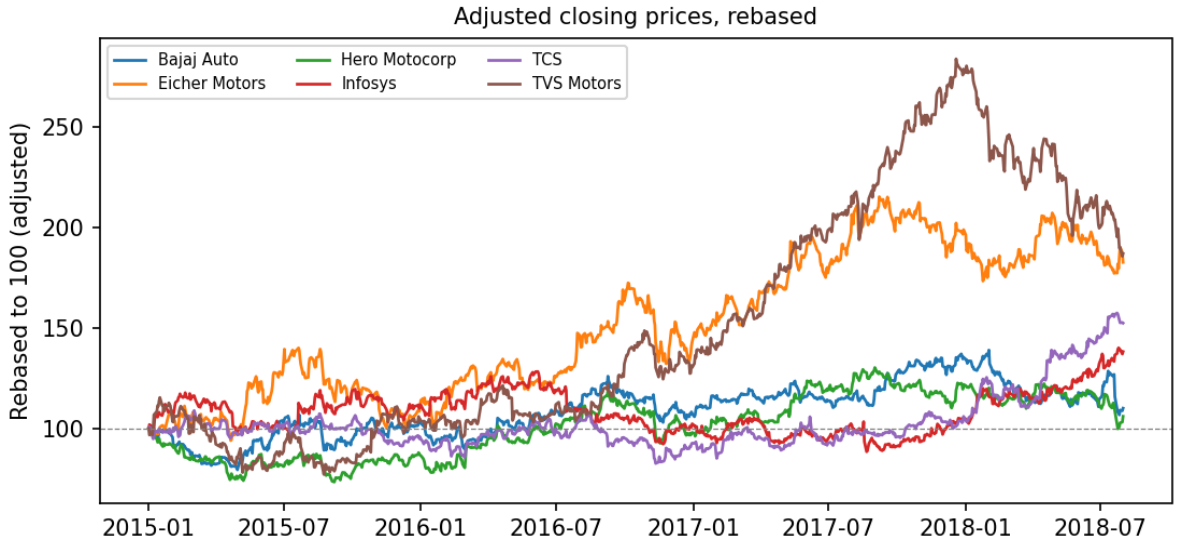
Stock Market Analysis in SQL: Insights Report

Six NSE stocks (Bajaj Auto, Eicher Motors, Hero Motocorp, Infosys, TCS, TVS Motors), 1 Jan 2015 to 31 Jul 2018, 889 trading days each. Method: 20-day vs 50-day moving-average golden cross (Buy when the 20-day average crosses above the 50-day, Sell when it crosses below). All numbers below come from my own MySQL 8 queries in *stock_market_analysis.sql*; none were copied from the course deck without checking.

1. Headline findings

• **Two price events in the data are not real price moves.** Infosys on **15 Jun 2015** (991.10, -49.9% in a day) and TCS on **31 May 2018** (1,744.80, -50.4%) are **1:1 bonus issues**: shares doubled, price halved, no value lost. I divided all earlier prices by 2 for each stock (task 13).
• **The winner/loser list flips for two stocks.** On raw prices TCS is -23.8% and Infosys -30.9%. Adjusted, TCS is **+52.4%** and Infosys **+38.2%**. Both are winners, not losers.
• **The bonus cliffs created false signals.** TCS's last raw signal, a Sell on 5 Jun 2018, exists only because of the cliff; adjusted, TCS's latest signal is a **Buy (20 Apr 2018)**. Infosys's raw Buy on 18 Aug 2015 was also created by the cliff, and adjusted prices reveal three signals the cliff hid (1 Jul, 8 Jul, 27 Jul 2015).
• **Corrected view:** latest signals are Buy for Bajaj Auto, Infosys and TCS, and Sell for Eicher Motors, Hero Motocorp and TVS Motors. The deck had TCS as a Sell.

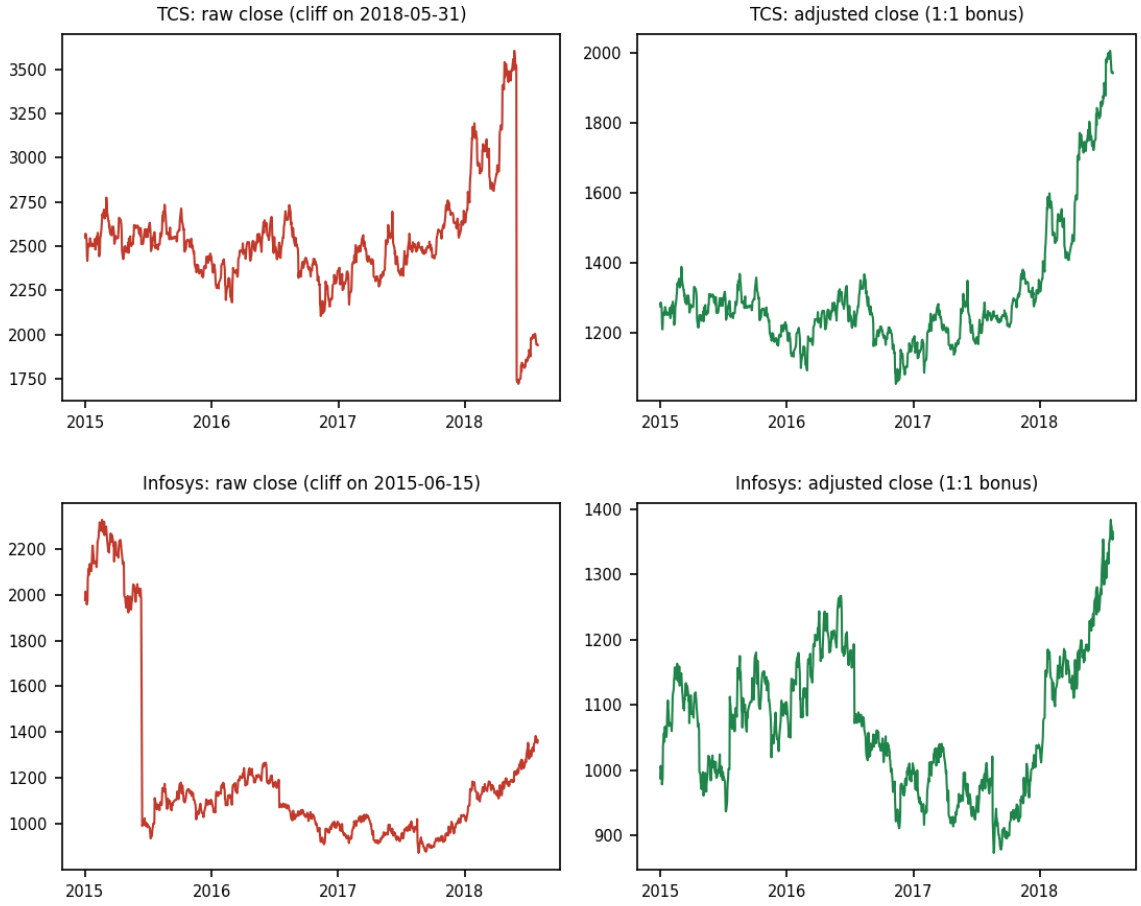


Stock	Raw % change	Adjusted % change	Buys	Sells	Latest signal (adjusted)	Close vs 50-day avg on 31 Jul
Bajaj Auto	+10.0%	+10.0%	12	11	Buy (2018-06-21)	-5.8%
Eicher Motors	+82.6%	+82.6%	6	7	Sell (2018-06-06)	-3.4%
Hero Motocorp	+6.0%	+6.0%	9	9	Sell (2018-05-22)	-6.2%
Infosys	-30.9%	+38.2%	10	10	Buy (2018-05-07)	+6.2%
TCS	-23.8%	+52.4%	12	12	Buy (2018-04-20)	+4.4%
TVS Motors	+86.9%	+86.9%	8	8	Sell (2018-05-17)	-9.3%

Buys/Sells are on adjusted prices; on raw prices Infosys shows 9/9 and TCS 12/13. Source: tasks 10, 11, 13.

2. Data quality: what the SQL checks found

• **Missing values (task 4):** A NULL count over every column of all six tables shows NULLs only in *deliverable_qty* (and *pct_deli_qty* on the same row), exactly one row per stock; I left them as NULL rather than filling with 0 or an average. *deliverable_qty* is NULL on one date per stock, and only on two distinct dates: 2015-12-09 (Eicher, Hero, TCS, TVS) and 2017-08-31 (Bajaj, Infosys). Because several unrelated companies share the same date, this looks like a gap in the exchange's reporting, not a company event. Closing prices are complete, so the signals are unaffected.
• **Tables line up (task 6):** all six tables share the same 889 dates, so the inner-join master table has 889 rows and no NULLs.
• **Worst day per stock (task 12):** Bajaj Auto -9.1% (24 Aug 2015), Eicher -9.3% (24 Aug 2015), TVS -9.7% (3 May 2016) and Hero -6.2% (23 Jul 2018) are in the normal range for a bad day (24 Aug 2015 was a broad market sell-off). Infosys and TCS show about -50%, which no profitable large company loses in a day without headlines. I checked outside sources: an NSE circular (23 May 2018) gives TCS a 1:1 bonus with ex-date 31 May 2018, and press coverage gives Infosys a 1:1 bonus with ex-date 15 Jun 2015 (record date 17 Jun 2015). Both match the cliff dates exactly.



How I handled it: for each stock I divided every close before the event date by 2 and left the event date and later unchanged (task 13). The stretch task rebuilt the moving averages and signals on adjusted prices. Caveat: adjusting for the bonus only fixes splits and bonuses; I did not adjust for dividends because the data has none.

3. Stock-by-stock insights (claim, evidence, caveat)

Stock	Bajaj Auto
Claim	Modest gain (+10.0%) with many signals; latest signal Buy (21 Jun 2018), but the price has since slipped below both averages.
Evidence	12 Buys, 11 Sells. 11 completed Buy-to-Sell round trips; 36% were profitable, average +0.4% per trip, compounded +2%; 5 lasted 20 trading days or fewer (average -4.1%). On 31 Jul the close (2,700.70) was 5.8% under the 50-day average (2,866.9) and the 20-day was 2,918.51.
Caveat	Following the signals compounded to only about +2% versus +10% for simply holding, with 11 round trips; a 'Buy' from six weeks ago is stale. Brokerage costs would reduce the 11 round trips further.

Stock	Eicher Motors
Claim	Second-best long-run performer (+82.6%, just behind TVS), but the latest signal is a Sell (6 Jun 2018).
Evidence	6 Buys, 7 Sells. 6 completed Buy-to-Sell round trips; 100% were profitable, average +10.2% per trip, compounded +76%; 0 lasted 20 trading days or fewer. On 31 Jul the 20-day average (27,757) sat below the 50-day (28,793).
Caveat	Every completed trip made money, but only 6 trips in 3.6 years is a tiny sample. A Sell here means 'momentum has cooled', not that the long-run uptrend has ended, so the long-run trend and the signal disagree and I would not act on the Sell alone.

Stock	Hero Motocorp
Claim	Flat (+6.0%) with a weak signal record; latest signal Sell (22 May 2018) agrees with the recent weakness.
Evidence	9 Buys, 9 Sells. 9 completed Buy-to-Sell round trips; 33% were profitable, average -1.7% per trip, compounded -16%; 1 lasted 20 trading days or fewer (average -6.5%). 31 Jul: 20-day 3,388 below 50-day 3,512.
Caveat	Sideways stocks are where crossovers work worst: the averages keep crossing without a real trend, so most trips lost money.

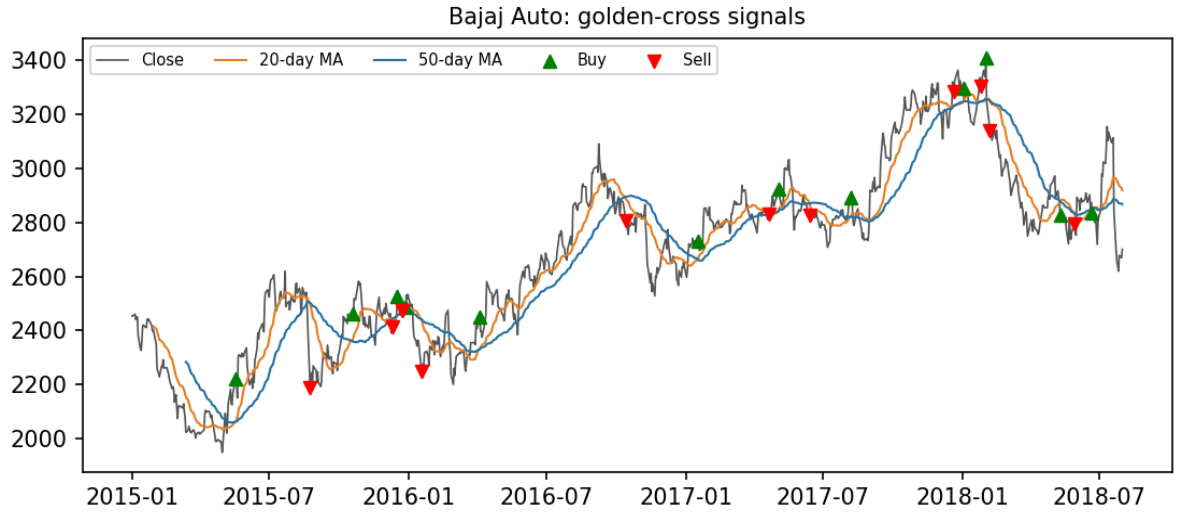
Stock	Infosys
Claim	After the bonus adjustment Infosys gained +38.2% (not -30.9%); latest signal Buy (7 May 2018) agrees with the long-run uptrend.
Evidence	10 Buys, 10 Sells on adjusted prices (9/9 raw). 9 completed Buy-to-Sell round trips; 33% were profitable, average -0.4% per trip, compounded -7%; 2 lasted 20 trading days or fewer (average -3.3%). 31 Jul: close 1,365 was 6.2% above the 50-day (1,285.5) and 20-day above 50-day.
Caveat	The raw series would have called Infosys a loser. Signals in Jun-Aug 2015 flip within weeks, so early history is noisy. The 2015 trip results are only as good as the factor-of-2 adjustment.

Stock	TCS
Claim	After the bonus adjustment TCS gained +52.4% (not -23.8%); latest signal is Buy (20 Apr 2018). The raw Sell on 5 Jun 2018 is an artefact.
Evidence	12 Buys, 12 Sells adjusted (12/13 raw). 11 completed Buy-to-Sell round trips; 18% were profitable, average -3.4% per trip, compounded -32%; 1 lasted 20 trading days or fewer (average -4.4%). 31 Jul: 20-day 1,954.8 above 50-day 1,858.6.
Caveat	TCS had the weakest signal record (few winning trips) despite the stronger adjusted uptrend than Infosys, a sign that this crossover rule lags a steadily trending stock. It also fired a Buy and a Sell on 13 and 14 Oct 2015, a one-day whipsaw.

Stock	TVS Motors
Claim	Biggest gain (+86.9%), yet the latest signal is a Sell (17 May 2018); the signal and the long-run trend disagree.
Evidence	8 Buys, 8 Sells. 8 completed Buy-to-Sell round trips; 38% were profitable, average +11.2% per trip, compounded +85%; 2 lasted 20 trading days or fewer (average -8.3%). 31 Jul: 20-day 560.0 below 50-day 570.3, close 517.45.
Caveat	Trip average is high because a few big trend trips carried it (win rate only 38%). The Sell may be catching a genuine pullback or a temporary dip; the data cannot tell them apart.

4. Whipsaws: when signals flip within weeks

A whipsaw is a Buy followed quickly by a Sell (or the reverse) with the price barely moving in between. In this data, short round trips of 20 trading days or fewer were **consistently losing**: Bajaj Auto (5 trips, average -4.1%), Hero (1, -6.5%), Infosys (2, -3.3%), TCS (1, -4.4%), TVS Motors (2, -8.3%); Eicher had none. Chasing every crossover would have cost roughly 3 to 8 percent per whipsaw before brokerage costs, which is why a short-lived signal deserves less trust than one that persists.



Bajaj Auto: the triangles cluster where the two averages tangle in sideways markets (2015-16 and 2017-18). Whipsaws come from sideways price action, not from any one stock being unusual.

5. Questions about the method

• **Moving averages only use the past.** A 50-day average needs 50 days to exist, and it reacts after price has already moved, so a golden cross confirms a trend rather than predicting one. The first Bajaj signal came on 18 May 2015, about four and a half months into the data.
• **What changes after task 13:** TCS goes from -23.8% to +52.4% (a swing of 76 percentage points), Infosys from -30.9% to +38.2% (69 points). TCS's latest signal changes from Sell to Buy, and it loses one Sell (13 to 12). Infosys gains one Buy and one Sell (9/9 to 10/10). The deck's Infosys figure of a 3% decrease does not match my query at all (raw -30.9%), and the deck's TCS start and end prices are swapped/misassigned (2,548.20 is the *first* close and 1,941.25 the last; the deck lists 2,454.1, which is Bajaj Auto's 2015 price). Eicher's rise is 82.6% to one decimal, the deck says 82.5%. The other stocks match.
• **What this data does not contain:** dividends (total return), corporate-action flags such as bonuses and splits (the source of the trap), news and results announcements, the broader market (Nifty), brokerage, taxes and slippage, and any risk measure such as volatility. 2018 has only 7 months of data.
• **Overall caveat:** four years and six stocks are too small a sample to prove any strategy works; these are descriptive results, not investment advice.